

Shijie Zhou

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EDUCATION

Technical University of Munich

Master of Science in Robotics, Cognition, Intelligence

Munich, Germany

Oct 2024 – Present

Changsha University of Science and Technology

Bachelor of Engineering in Electrical Engineering and Automation

Changsha, China

Sep 2020 – Jun 2024

RESEARCH INTERESTS

My research lies at the intersection of **robotics**, **embodied AI**, and **foundation models**. My long-term goal is to develop general-purpose robotic agents capable of robust perception, reasoning, and manipulation in open-world environments. I am currently interested in:

- **Multimodal Learning** for robot perception.
- **World Models** for prediction, reasoning, and planning.
- **Foundation Models for Robotics** for scalable and generalizable robot learning.

PUBLICATIONS

PREPRINTS

[P1] *How to Train a Surgeon? Benchmarking Generalist Agents in Surgical Scene Understanding*. G. Zhang, X. Han*, **S. Zhou***, A. Koukash*, G. Ghazaei, J. Gu, J. Wu, Y. Zhu, N. Navab, and V. Tresp. **NeurIPS 2026 (Under Review)**.

[P2] *PrivacyEvalClaw: Towards Deliberative Tool-Augmented Agent for LLM Privacy Jailbreak Evaluation*. K. K. Nakka, Y. Wang, **S. Zhou**, T. Nguyen, G. Gelardi, and X. Zhou. **EMNLP 2026 (Under Review)**.

CONFERENCE PROCEEDINGS

[C1] *SwarmAgentic: Towards Fully Automated Agentic System Generation via Swarm Intelligence*. Y. Zhang, C. Lin, S. Tang, H. Chen, **S. Zhou**, Y. Ma, and V. Tresp. **EMNLP (Main) 2025**. [Paper] [Code]

*Equal contribution.

RESEARCH EXPERIENCE

University of Cambridge

Research Assistant at *CV4DT Group*, advised by *Dr. Zhengyang Ling* and *Dr. Guangming Wang*

Cambridge, UK

Apr 2026 – Present

- Research Topic: Vision-Supervised System Identification for Soft Robotics.
- Developed a vision-supervised digital twin of a three-finger soft gripper in SoMo/PyBullet, using four calibrated RGB cameras and CoTracker3 to reconstruct 3D deformation trajectories for soft-body system identification.
- Identified a 2-axis soft-body dynamics model via Levenberg–Marquardt optimization and established a unified benchmark for systematic comparison of model-based and learning-based control strategies.

TU Munich

Research Assistant at *AIR Lab*, advised by *Mohammadhossein Malmir* and *Prof. Alois Knoll*

Munich, Germany

Oct 2025 – Present

- Research Topic: Agentic Hierarchical Reward Specification for Reinforcement Learning in Robotics.
- Proposed ReGoRL, a tool-augmented agentic framework based on hierarchical potential-based reward shaping for safety-critical robot reinforcement learning.

Huawei Munich Research Center

Research Intern at the *TTE Lab*, advised by *Dr. Krishna Kanth Nakka*

Munich, Germany

Oct 2025 – Apr 2026

- Research Topic: Privacy Alignment and Jailbreak Evaluation for Large Language Models.
- Conducted a data-centric study of privacy alignment by constructing controlled dataset variants and identifying predictive data signals that govern the privacy–over-refusal–utility tradeoff in LLMs.
- Proposed a tool-augmented agentic framework for LLM privacy jailbreak evaluation that reconciles conflicting guard and judge signals through adaptive, multi-tool investigation, improving agreement with human annotations on high-disagreement cases.

- *Research Topic: Agentic Multimodal Learning for Surgical Video Understanding.*
- Built an agentic evaluation framework for surgical multimodal understanding, integrating bounded retrieval, visual grounding, and trajectory-level analysis for tool-using MLLM agents.
- Proposed a PSO-inspired framework for automated agentic system generation that optimize agent functionality and collaboration through language-driven exploration, outperforming existing baselines in unconstrained tasks.

SELECTED PROJECTS

Foundation Model-based Representation Learning for Vision-Guided Robotic Ultrasound [\[Code\]](#)

Mar 2026 – Present

- Developed a geometry-aware real-to-sim pipeline that scales sparse robotic ultrasound acquisitions into large-scale synthetic datasets through CBCT reslicing, LC2 registration, paired ultrasound rendering, and physically valid trajectory generation.
- Benchmarked general, medical, and ultrasound foundation models for robotic ultrasound segmentation, evaluating the impact of geometry-aware synthetic data under limited real-data supervision.

LLM-assisted Hierarchical Reward Shaping for Offline-to-Online RL in Warehouse Navigation [\[Paper\]](#) [\[Code\]](#)

Oct 2025 – Mar 2026

- Built an offline-to-online RL pipeline for AGV navigation using Nav2 critic-based (A* + DWB) expert data and TD3-BC initialization with domain randomization, outperforming training from scratch in controlled evaluations.
- Developed an LLM-assisted Hierarchical Potential-based Reward Shaping (HPRS) framework for automated failure diagnosis and reward refinement, enabling systematic policy improvement.

Simulation-based Modeling and Control of Humanoid Robot [\[Code\]](#)

Apr 2025 – Jul 2025

- Built a ROS2–PyBullet–Pinocchio simulation and control stack for the TALOS humanoid, implementing floating-base dynamics, inverse dynamics control, and whole-body TSID for standing and single-leg balance.
- Integrated ZMP/DCM-based balance strategies with LIP-MPC footstep planning and swing-foot trajectory generation to achieve stable walking in simulation, with disturbance rejection experiments and quantitative evaluations.

SKILL SET

Programming: Python (PyTorch, TensorFlow, HuggingFace, NumPy, OpenCV), C++, MATLAB, Linux Shell, L^AT_EX

Simulation & Robotics Tools: ROS1/2, PyBullet, Genesis, Pinocchio, Git, Anaconda

Robots: Franka Research 3

Languages: Chinese, English, German (TestDaF: 18/20)